The Six Dimensions of Data Quality

In this train, we'll explore methods for scaling Apache SpaIn this train, we'll take a look at the six dimensions of data quality, providing you with a comprehensive understanding of each dimension and how they contribute to maintaining high data quality standards.

Learning Objectives

By the end of this train, you should be able to:

- Define the six dimensions of data quality.

The Six Dimensions of Data Quality

The six dimensions of data quality are generalised attributes that can be used to measure data quality. These dimensions provide a framework within which you can explore data and understand where pitfalls may be, as well as how to address data quality problems.



The six dimensions of data quality.

In some cases, specific dimensions will not apply to a dataset and, therefore, may not be useful, however, it is important to go through all six dimensions for each use case. It is, therefore, useful to have a standard approach to apply to new datasets. If there are additional requirements, we can build these on top of the six dimensions.

An Overview of The Six Dimensions of Data Quality

Each of the six dimensions should give you an idea of how to assess your data's quality.

Accuracy

The accuracy of data is the degree to which the data represent a real-world event or object.

Sometimes IoT devices record data measurements that are incorrect but may look like a real value. An example of this is monitors that measure the level of fluid in a tank or pipe. These monitors often use an ultrasonic pulse that bounces off the fluid to measure the level. However, if the sensor is underwater or obstructed, it can give skewed readings. Another example is an API sending incorrect data or ingesting the data from the source incorrectly, such as a rounding error.

Accuracy issues can occur at the field level (one incorrect entry) or the row level (multiple incorrect rows).

Another issue that may occur is default values. A typical example of this is where a logger sends back a 0 instead of a null value, which can greatly skew any attempts at modelling. This is where it is instrumental to employ domain knowledge when assessing a dataset.

- *Measured by*: how correct a data point is relative to a real-world event
- *Units*: specific case-related constraints
- *Related to*: validity and completeness

Consistency

Consistency is the absence of difference when comparing two or more representations of something against a reference. If data are recorded or captured in multiple places, consistency becomes very important. One cannot have the same data point recorded in various ways.

Data entries that refer to the same record or entity have to be consistent across all of the entries. An example of a consistency error is measuring data in different units but using the same column. For example, temperature data that are recorded in Kelvin and degrees Celsius but in the same column. This could greatly impact data users if they are unaware of what normal temperatures should look like.

This is not just within a single table. It becomes more important if you are dealing with relational data, in which case the mappings between tables and systems must be consistent. If not, the relationships will be completely lost between the tables and referential integrity compromised.

- *Measured by*: analysis of pattern and/or value frequency
- *Units*: percentage
- *Related to*: accuracy, validity, and uniqueness

Timeliness

Timeliness is the degree to which data represent reality from the required point in time. Timeliness expects that the data within your dataset is sufficiently up to date. What are the delays between an event happening and the data point being recorded?

If you are trying to answer questions that relate to recent problems, having timely data is extremely important. For example, you cannot use current flight patterns to model how many aeroplanes will be needed by a large aeronautics company in the next five to ten years.

Similarly, when answering questions that require real-time answers (for example, predicting when a pipe will burst in a manufacturing plant), you have to be set up to receive real-time data from sensors and loggers.

An example of untimely data causing problems is where a temperature measuring device on a nuclear reactor only communicates its reading two hours after the temperature is recorded. If the temperature exceeds the safety threshold, it could lead to a nuclear meltdown in the reactor. Therefore, it is crucial that the data are sent as soon as it is recorded and that the data pipelines can communicate this data point as fast as possible.

Another important point is that timeliness is sometimes determined by the source system, which may only synchronise once a day. In this case, we may have a fast method of ingesting the data and displaying it, but we are limited by the source system.

- *Measured by*: time difference
- *Units*: time

- *Related to*: accuracy, because it will decay as time progresses

Validity

The validity of a dataset is specific to a certain field. In other words, data is valid if it conforms to the syntax (format, type, range) of its definition. Each field will have a property that makes it valid, such as an "@" symbol for an email address.

Certain values within a field may have criteria required to make it valid, for example, numerical columns cannot contain alphabetical characters, which can be the result of incorrectly parsed scientific notation. This can be more difficult to determine in strings, in which case you may have to check using regex. This could be the inclusion of a symbol, as mentioned above, or it could be that an integer should not exceed certain limits. An example of this is in a gas separation process that uses extremely cold temperatures. If the system records temperatures below -273 degrees Celsius, which is absolute zero, the data would not be valid.

- *Measured by*: comparison between the data and the metadata or documentation for the data item
- *Units*: percentage of data items deemed valid or invalid
- *Related to*: accuracy, completeness, consistency, and uniqueness

Completeness

The completeness of data relates to how many values may be missing in your dataset.

The degree of completeness can be measured by seeing how many theoretical data points should be present, based on timestamps. Once a theoretical maximum of how complete a dataset should be is established, you can determine how complete the dataset is based upon the ratio of missing records to theoretical completeness.

An example of incomplete data is if you are receiving pump flow readings every two minutes, but you do not have 720 data points per day. If you have 600 data points associated with a specific timestamp, it means that you are missing 120 data points/timestamps. Being aware of this is particularly important in cases where you are aggregating the data to a less frequent interval, because you can still get an hourly mean with less than 30 two-minute data points but it will not be as accurate as with all 30 data points.

The questions one could ask to determine the completeness of a dataset are, 'Does the dataset have missing values, or if it is time-series data, does it have time period gaps? Has a bias been introduced that may change your assumptions or affect your results?'

Completeness issues can occur at the field level (one entry missing) or at the row level (gaps within the dataset). This can also happen at the field level with entire fields being empty or more than 80% of a field's data being missing. Another issue that may occur is default values. A typical example of this is where a logger sends back a 0 instead of a null value, which can greatly skew any attempts at modelling. This is where it is instrumental to employ domain knowledge when assessing a dataset.

- *Measured by*: a measure of the absence of blank (null) values or the presence of non-blank values
- *Units*: percentage
- *Related to*: validity and accuracy

Uniqueness

This dimension relates to having a real-world object or event represented only once in a particular dataset. The same object cannot be duplicated. In other words, uniqueness specifies that nothing will be recorded more than once based upon how that thing is identified. It is the inverse of an assessment of the level of duplication.

Each entry within the datasets should only relate to one single event which has occurred and, thus, should not be duplicated. This is largely mediated by having the appropriate primary key, which is ensured if you stick to the requirements of a good primary key. All fields in the tables should be non-transitively dependent on the primary key. As such, deduplication of the dataset may be required.

This dimension of data quality can lead to very frustrating situations. An example is having a replication of a set of asset IDs that relate to pumps. Each asset has a flow measurement device, and each has a theoretical maximum that it can pump. If an asset's name is recorded as both ADZZ-001 and adzz-001, data could be recorded multiple times for the same pump. The theoretical maximum pumping limit could also be reached for an asset that then kicks off a process to alert users to the problem. This could lead to users receiving double the number of alarms or alerts and potentially causing more stress than is necessary.

- *Measured by*: analysis of the number of things assessed in the “real world” compared to the number of records of things in the dataset – it requires a reference dataset which is the ground truth
- *Units*: percentage
- *Related to*: consistency

Conclusion

Ensuring high data quality is crucial for producing reliable insights and powering accurate decision-making. By evaluating your data across the six dimensions—accuracy, consistency, timeliness, validity, completeness, and uniqueness—you can identify potential issues and implement improvements. Each of these dimensions offers a unique lens through which to assess the integrity and reliability of your dataset, enabling better outcomes in analytics, modeling, and real-world applications.

▼Key Takeaways ✖

- **Takeaway 1:** Accuracy, measures how closely data reflects real-world events; inaccuracies can arise from faulty sensors, API errors, or default values.
- **Takeaway 2:** Consistency, ensures data remains uniform across sources; inconsistencies like mixed units or mismatched relationships can break data logic.
- **Takeaway 3:** Timeliness, assesses how current the data is relative to events; delays in data capture or source system syncs can reduce its usefulness.
- **Takeaway 4:** Validity, checks if data conforms to expected formats or rules; invalid data often violates type, range, or format expectations.
- **Takeaway 5:** Completeness, evaluates if all expected data is present; missing values or default placeholders like zeroes can skew analysis.
- **Takeaway 6:** Uniqueness, confirms each real-world entity is represented once; duplicates can distort metrics and lead to inaccurate conclusions.